

Stock Market Analysis Dashboard Report

1. Introduction

The Stock Market Dashboard is a web-based application designed to provide users with real-time and historical stock data in an interactive interface.

2. Objectives

- Develop an interactive financial dashboard using Streamlit
- Integrate real-time stock data using an external API
- Visualize stock trends through charts
- Provide insights such as price trends and moving averages

3. Technologies Used

Python, Streamlit, yfinance API, Pandas, Streamlit Charts

4. System Features

- Stock symbol search
- Time range selection
- Real-time metrics (price, high, low)
- Interactive charts
- Historical data table
- Moving averages (MA5, MA10)
- Error handling

5. System Workflow

User input → Validation → Data Fetching → Processing → Visualization → Display

6. Implementation Highlights

- Used yfinance for real-time data
- Applied rolling averages for trend analysis
- Designed responsive UI with Streamlit
- Implemented exception handling

7. Results

The system displays accurate stock data, clear visualizations, and an interactive UI.

8. Conclusion

This project demonstrates integration of APIs, data analysis, and web development, forming a foundation for advanced analytics systems.

System Diagrams

